

Scalability & Future Plan

Date	15 July 2026
Team ID	XXXXXX
Project Name	Human Development Index (HDI) Predictor
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Uses Linear Regression only	Prediction accuracy can be improved with advanced models	High
2	Local CSV dataset	Cannot update data automatically	Medium
3	Runs on local Flask server	Not accessible over the Internet	High

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports local users	Deploy on cloud using Render or Azure
2	Data Storage	CSV dataset	Integrate SQL or MongoDB database
3	Performance	Linear Regression model	Use XGBoost or Random Forest for better accuracy
4	Security	Basic Flask application	Add authentication, HTTPS, and input validation

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
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Phase 1	Improve UI/UX	1 Month	Better user experience
Phase 2	Add Country Search Feature	2 Months	Easier prediction process
Phase 3	Deploy on Cloud	3 Months	Public accessibility
Phase 4	Dashboard & Analytics	4 Months	Interactive visual insights